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JEE MAIN PHYSICS DPP: OBLIQUE PROJECTILE MOTION

Topic: Oblique Projectile: Max Height, Time of Flight, Range & Trajectory | **Time: 45 Min** | **Max Marks: 80**

Marking Scheme: **+4, -1** | Designed & Curated by: **Syed Mazhar Ali**

Q1. A projectile is fired with an initial speed u at an angle θ with the horizontal. The total time of flight T is:

(a) $\dfrac{u\sin\theta}{g}$

(b) $\dfrac{2u\sin\theta}{g}$

(c) $\dfrac{u\cos\theta}{g}$

(d) $\dfrac{2u\tan\theta}{g}$

Q2. If the horizontal range of a projectile is four times its maximum height ($R = 4H$), the angle of projection θ is:

(a) $30°$

(b) $60°$

(c) $45°$

(d) $90°$

Q3. Two projectiles are thrown with the same initial speed u at complementary angles θ and $(90° - \theta)$. If H_1 and H_2 are their maximum heights and R is their common range, then R is equal to:

(a) $2\sqrt{H_1H_2}$

(b) $H_1 + H_2$

(c) $\sqrt{H_1H_2}$

(d) $4\sqrt{H_1H_2}$

Q4. In terms of the horizontal range R, the standard equation of trajectory of an oblique projectile is written as:

(a) $y = x\tan\theta\left(1 - \dfrac{x}{R}\right)$

(b) $y = x\cot\theta\left(1 - \dfrac{x}{R}\right)$

(c) $y = x\tan\theta\left(1 + \dfrac{x}{R}\right)$

(d) $y = \dfrac{x}{R}\tan\theta$

Q5. A particle is projected with velocity u at an angle θ to the horizontal. The radius of curvature of its trajectory at the highest point is:

(a) $\dfrac{u^2}{g}$

(b) $\dfrac{u^2\cos^2\theta}{g}$

(c) $\dfrac{u^2\sin^2\theta}{g}$

(d) $\dfrac{u^2}{g\cos\theta}$

Q6. The kinetic energy of a projectile at its highest point is found to be half of its initial kinetic energy at launch. The angle of projection is:

(a) $30°$

(b) $60°$

(c) $45°$

(d) $75°$

Q7. For two projectiles launched with the same speed at complementary angles θ and $(90° - \theta)$, if T_1 and T_2 are their respective times of flight and R is their horizontal range, then T_1T_2 is equal to:

(a) $\dfrac{R}{g}$

(b) $\dfrac{4R}{g}$

(c) $\dfrac{R}{2g}$

(d) $\dfrac{2R}{g}$

Q8. The equation of trajectory of a projectile is given by $y = \sqrt{3}x - 5x^2$ (where x, y are in meters and $g = 10\ \text{m/s}^2$). The initial speed of projection u is:

(a) $2\sqrt{2}$ m/s

(b) 4 m/s

(c) 2 m/s

(d) $\sqrt{3}$ m/s

Q9. A projectile is launched from ground level with speed u at angle θ. The angle α made by the velocity vector with the horizontal at time t is given by:

(a) $\tan\alpha = \dfrac{u\sin\theta}{gt}$

(b) $\tan\alpha = \tan\theta - \dfrac{gt}{u\cos\theta}$

(c) $\tan\alpha = \tan\theta + \dfrac{gt}{u\cos\theta}$

(d) $\tan\alpha = \dfrac{u\cos\theta - gt}{u\sin\theta}$

Q10. A ball is projected from the ground with speed u at an angle θ. The magnitude of change in momentum of the ball over the entire time of flight is:

(a) $2mu$

(b) $2mu\cos\theta$

(c) $2mu\sin\theta$

(d) $mu\sin\theta$

Q11. A particle is projected from ground level with speed u at angle θ. The average velocity of the projectile between the instant of projection and the instant it lands back on the ground is:

(a) u
(b) $u \sin\theta$
(c) $\dfrac{u}{2}$
(d) $u \cos\theta$

Q12. If R is the maximum horizontal range achievable for a projectile with a given launch speed, the maximum height attained during that optimal launch is:

(a) $\dfrac{R}{4}$
(b) $\dfrac{R}{2}$
(c) R
(d) $\dfrac{R}{8}$

Q13. A stone is projected with velocity u at angle θ. The time t after launch at which its velocity vector becomes perpendicular to the initial velocity vector $\vec{u}$ is:

(a) $\dfrac{u}{g\sin\theta}$
(b) $\dfrac{u}{g\sin\theta}$ (provided $\theta > 45°$)
(c) $\dfrac{u\cos\theta}{g}$
(d) $\dfrac{2u}{g\cos\theta}$

Q14. A projectile has a maximum height $H = 20$ m and range $R = 80$ m. The initial velocity of projection is: (Take $g = 10$ m/s^2)

(a) 20 m/s
(b) 40 m/s
(c) $20\sqrt{2}$ m/s
(d) $10\sqrt{5}$ m/s

Q15. A ball thrown at angle θ passes over two vertical poles of equal height h situated at distances d_1 and d_2 from the point of projection. The horizontal range R of the projectile is:

(a) $\sqrt{d_1 d_2}$
(b) $\dfrac{d_1 + d_2}{2}$
(c) $\dfrac{2d_1 d_2}{d_1 + d_2}$
(d) $d_1 + d_2$

Q16. In Q15, the height h of the two poles in terms of d_1, d_2, and launch angle θ is:

(a) $\dfrac{d_1 d_2 \tan\theta}{d_1 + d_2}$
(b) $\dfrac{(d_1 + d_2)\tan\theta}{d_1 d_2}$
(c) $(d_1 + d_2)\tan\theta$
(d) $\sqrt{d_1 d_2}\tan\theta$

Q17. The speed of a projectile at its highest point is $\dfrac{\sqrt{3}}{2}u$, where u is its launch speed. The horizontal range of the projectile is:

(a) $\dfrac{u^2}{2g}$
(b) $\dfrac{\sqrt{3}u^2}{2g}$
(c) $\dfrac{u^2}{g}$
(d) $\dfrac{\sqrt{3}u^2}{g}$

Q18. A body is projected at an angle $\theta = 30°$ with kinetic energy K. Its potential energy at the highest point (taking launch level as $PE = 0$) is:

(a) $\dfrac{3K}{4}$
(b) $\dfrac{K}{2}$
(c) $\dfrac{K}{4}$
(d) $\dfrac{K}{8}$

Q19. A particle moves along a parabolic path $y = ax - bx^2$ under gravity where x, y are horizontal and vertical displacements. The angle of projection θ and initial speed u are:

(a) $\tan\theta = b,\ u = \sqrt{\dfrac{g(1+a^2)}{2b}}$
(b) $\tan\theta = a,\ u = \sqrt{\dfrac{g}{2b}}$
(c) $\tan\theta = \dfrac{a}{b},\ u = \sqrt{\dfrac{g}{2ab}}$
(d) $\tan\theta = a,\ u = \sqrt{\dfrac{g(1+a^2)}{2b}}$

Q20. The time of flight T and horizontal range R of an oblique projectile are related by $gT^2 = 2R\tan\theta$. If the projection speed is doubled keeping θ constant, the value of gT^2/R will:

(a) Remain unchanged $(2\tan\theta)$
(b) Double
(c) Become 4 times
(d) Halve

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	**6**	(c)	**11**	(d)	**16**	(a)
2	(c)	**7**	(d)	**12**	(a)	**17**	(b)
3	(d)	**8**	(a)	**13**	(b)	**18**	(c)
4	(a)	**9**	(b)	**14**	(c)	**19**	(d)
5	(b)	**10**	(c)	**15**	(d)	**20**	(a)

Detailed Solutions

Sol 1. Correct Option: (b)
Setting vertical displacement $y(T) = 0$:

$$(u\sin\theta)T - \frac{1}{2}gT^2 = 0 \implies T = \frac{2u\sin\theta}{g}$$

Sol 2. Correct Option: (c)
$R = \frac{u^2\sin 2\theta}{g} = \frac{2u^2\sin\theta\cos\theta}{g}$, and $H = \frac{u^2\sin^2\theta}{2g}$.
Given $R = 4H$:

$$\frac{2u^2\sin\theta\cos\theta}{g} = 4\left(\frac{u^2\sin^2\theta}{2g}\right)$$

$$\sin 2\theta = 2\sin^2\theta \implies \tan\theta = 1 \implies \theta = 45^\circ$$

Sol 3. Correct Option: (d)
$H_1 = \frac{u^2\sin^2\theta}{2g}$, $H_2 = \frac{u^2\cos^2\theta}{2g}$.
Multiplying H_1H_2:

$$H_1H_2 = \frac{(u^2\sin 2\theta)^2}{16g^2} = \frac{R^2}{16}$$

$$R = 4\sqrt{H_1H_2}$$

Sol 4. Correct Option: (a)
$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$.
Factoring out $x\tan\theta$ with $R = \frac{2u^2\sin\theta\cos\theta}{g}$ yields:

$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$

Sol 5. Correct Option: (b)
At the highest point, speed is purely horizontal: $v = u\cos\theta$.
Normal acceleration is $a_n = g$.
Radius of curvature:

$$R_c = \frac{v^2}{a_n} = \frac{u^2\cos^2\theta}{g}$$

Sol 6. Correct Option: (c)
At the highest point, kinetic energy is:

$$K_{\text{top}} = \frac{1}{2}m(u\cos\theta)^2 = K_0\cos^2\theta$$

Given $K_{\text{top}} = \frac{1}{2}K_0 \implies \cos^2\theta = \frac{1}{2} \implies \theta = 45^\circ$.

Sol 7. Correct Option: (d)
$T_1 = \frac{2u\sin\theta}{g}$, $T_2 = \frac{2u\cos\theta}{g}$.
Multiplying:

$$T_1T_2 = \frac{4u^2\sin\theta\cos\theta}{g^2} = \frac{2}{g}\left(\frac{u^2\sin 2\theta}{g}\right) = \frac{2R}{g}$$

Sol 8. Correct Option: (a)
Comparing $y = \sqrt{3}x - 5x^2$ with $y = x\tan\theta - \frac{g}{2u^2\cos^2\theta}x^2$:

- $\tan\theta = \sqrt{3} \implies \theta = 60^\circ \implies \cos\theta = \frac{1}{2}$.
- $\frac{10}{2u^2(1/4)} = 5 \implies \frac{20}{u^2} = 5 \implies u^2 = 8 \implies u = 2\sqrt{2}$ m/s.

Sol 9. Correct Option: (b)
$v_x = u\cos\theta$, $v_y = u\sin\theta - gt$.
Angle of velocity with horizontal:

$$\tan\alpha = \frac{v_y}{v_x} = \frac{u\sin\theta - gt}{u\cos\theta} = \tan\theta - \frac{gt}{u\cos\theta}$$

Sol 10. Correct Option: (c)
$\vec{p}_i = m(u\cos\theta\hat{i} + u\sin\theta\hat{j})$.
$\vec{p}_f = m(u\cos\theta\hat{i} - u\sin\theta\hat{j})$.
Change in momentum:

$$\Delta\vec{p} = \vec{p}_f - \vec{p}_i = -2mu\sin\theta\hat{j}$$

$$|\Delta\vec{p}| = 2mu\sin\theta$$

Sol 11. Correct Option: (d)
Total displacement is horizontal range $\vec{S} = R\hat{i}$.
Total time is $T = \frac{2u\sin\theta}{g}$.
Average velocity:

$$\vec{v}_{\text{avg}} = \frac{\vec{R}}{T} = \frac{(u\cos\theta)T\hat{i}}{T} = u\cos\theta\hat{i}$$

Sol 12. Correct Option: (a)

Maximum range occurs at $\theta = 45°$: $R_{\max} = \dfrac{u^2}{g}$.
Maximum height at $\theta = 45°$:

$$H = \frac{u^2 \sin^2(45°)}{2g} = \frac{u^2(1/2)}{2g} = \frac{u^2}{4g} = \frac{R_{\max}}{4}$$

Sol 13. Correct Option: (b)

$\vec{u} = u\cos\theta\hat{i} + u\sin\theta\hat{j}$.
$\vec{v}(t) = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}$.
Perpendicular condition $\vec{u}\cdot\vec{v} = 0$:

$$u^2\cos^2\theta + u^2\sin^2\theta - ugt\sin\theta = 0$$
$$u^2 - ugt\sin\theta = 0 \implies t = \frac{u}{g\sin\theta}$$

Sol 14. Correct Option: (c)

$\dfrac{R}{H} = \dfrac{4}{\tan\theta} \implies \dfrac{80}{20} = \dfrac{4}{\tan\theta} \implies \tan\theta = 1 \implies \theta = 45°$.
$H = \dfrac{u^2\sin^2(45°)}{2g} = 20 \implies \dfrac{u^2(1/2)}{20} = 20 \implies u^2 = 800$.

$$u = \sqrt{800} = 20\sqrt{2}\ \text{m/s}$$

Sol 15. Correct Option: (d)

Using trajectory equation $y = x\tan\theta\left(1 - \dfrac{x}{R}\right)$ at $x = d_1$ and $x = d_2$ for same height h:
$h = d_1\tan\theta\left(1 - \frac{d_1}{R}\right) = d_2\tan\theta\left(1 - \frac{d_2}{R}\right)$.
Equating and simplifying yields $R = d_1 + d_2$.

Sol 16. Correct Option: (a)

Substituting $R = d_1 + d_2$ into $h = d_1\tan\theta\left(1 - \dfrac{d_1}{d_1 + d_2}\right)$:

$$h = d_1\tan\theta\left(\frac{d_2}{d_1 + d_2}\right) = \frac{d_1 d_2\tan\theta}{d_1 + d_2}$$

Sol 17. Correct Option: (b)

At the highest point, speed $v = u\cos\theta = \dfrac{\sqrt{3}}{2}u \implies \cos\theta = \dfrac{\sqrt{3}}{2} \implies \theta = 30°$.
Range:

$$R = \frac{u^2\sin(2\times 30°)}{g} = \frac{u^2\sin(60°)}{g} = \frac{\sqrt{3}u^2}{2g}$$

Sol 18. Correct Option: (c)

Initial KE: $K = \frac{1}{2}mu^2$.
At highest point, KE is $K_{\text{top}} = \frac{1}{2}m(u\cos 30°)^2 = K\left(\frac{3}{4}\right)$.
By conservation of energy:

$$U_{\text{top}} = K - K_{\text{top}} = K - \frac{3K}{4} = \frac{K}{4}$$

Sol 19. Correct Option: (d)

Comparing $y = ax - bx^2$ with $y = x\tan\theta - \dfrac{g(1+\tan^2\theta)}{2u^2}x^2$:

- $\tan\theta = a$.
- $b = \dfrac{g(1+a^2)}{2u^2} \implies u = \sqrt{\dfrac{g(1+a^2)}{2b}}$.

Sol 20. Correct Option: (a)

$T = \dfrac{2u\sin\theta}{g} \implies T^2 = \dfrac{4u^2\sin^2\theta}{g^2}$.
$R = \dfrac{2u^2\sin\theta\cos\theta}{g}$.

$$\frac{gT^2}{R} = \frac{g\cdot\frac{4u^2\sin^2\theta}{g^2}}{\frac{2u^2\sin\theta\cos\theta}{g}} = 2\tan\theta$$

Since θ is unchanged, the ratio remains constant $(2\tan\theta)$.